

**Multiple Choice:**

- "If objects 1 and 2 are each in thermal equilibrium with object 3, then objects 1 and 2 are in thermal equilibrium with each other." This statement refers to which law of Thermodynamics?
A. Zeroth Law of Thermodynamics
B. First Law of Thermodynamics
C. Second Law of Thermodynamics
D. Third Law of Thermodynamics
- "Energy (in the form of heat) is neither created nor destroyed in any thermodynamic system." This statement refers to which law of Thermodynamics?
A. Zeroth Law of Thermodynamics
B. First Law of Thermodynamics
C. Second Law of Thermodynamics
D. Third Law of Thermodynamics
- Through a series of thermodynamic processes, the internal energy of a sample of confined gas is increased by 560 J. If the net amount of work done on the sample by its surroundings is 320 J, how much heat was transferred between the gas and its environment?
A. 240 J absorbed
B. 240 J dissipated
C. 880 J absorbed
D. 880 J dissipated
- "The total amount of disorder the total entropy of a system plus its surroundings will never decrease." This statement refers to which law of Thermodynamics?
A. Zeroth Law of Thermodynamics
B. First Law of Thermodynamics
C. Second Law of Thermodynamics
D. Third Law of Thermodynamics
- A heat engine draws 800 J of heat from its high-temperature source and discards 600 J of exhaust heat into its cold-temperature reservoir each cycle. How much work does this engine perform per cycle, and what is its thermal efficiency?
A. 15%
B. 20%
C. 25%
D. 40%
- What happens to the rms speed of the molecules in a sample of helium gas if the temperature is increased from -73°C to 527°C ?
A. Increases by a factor of 0.5
B. Increases by a factor of 2
C. Increases by a factor of 4
D. Increases by a factor of sqrt 2
- What happens to the pressure of a sample of helium gas if the temperature is increased from 200 K to 800 K, with no change in volume?
A. P will decrease by a factor of 4
B. P will increase by a factor of 4
C. P will not change
D. P will increase by a factor fo 2
- What happens to the pressure of a sample of helium gas if the volume is reduced from 6 liters to 3 liters, with no change in temperature?
A. P will decrease by a factor of 4
B. P will increase by a factor of 4
C. P will not change
D. P will increase by a factor fo 2
- Gold has a specific heat of $130 \text{ J/kg}\cdot^{\circ}\text{C}$, and wood has a specific heat of $1,800 \text{ J/kg}\cdot^{\circ}\text{C}$. If a piece of gold and a piece of wood, each of mass 0.1 kg, both absorb 2,340 J of heat, by how much will the temperature of gold and wood, respectively, rise?
A. 90°C and 24°C
B. 180°C and 13°C
C. 150°C and 11°C
D. 180°C and 25°C
- The heat of fusion for water is 334 kJ/kg . How much thermal energy is required to completely melt a 100-gram ice cube?
A. 3.34 kJ
B. 33.4 kJ
C. 51.1 kJ
D. 5.11 kJ
- A block of Teflon of mass 0.05 kg at a temperature of 430°C is placed in a beaker containing 0.5 kg of water (10 times the mass of the Teflon block) at room temperature, 20°C . After a while, the Teflon and water reach thermal equilibrium (they're at the same temperature). If the specific heat of water is 4 times the specific heat of Teflon, what's the common final temperature of the Teflon and the water?
A. 15°C
B. 22°C
C. 30°C
D. 18°C
- A beaker contains 0.1 kg of water, initially at room temperature (20°C). If the specific heat of water is $4.2 \text{ J/kg} \times ^{\circ}\text{C}$ and the latent heat of vaporization is $2,300 \text{ kJ/kg}$, how much thermal energy would the water need to absorb to turn completely to steam?
A. 230 kJ
B. 33.6 kJ
C. 264 kJ
D. 291 kJ
- A steel beam used in the construction of a bridge has a length of 30.0 m when the temperature is 15°C . On a very hot day, when the temperature is 35°C , how long will the beam be? (The coefficient of linear expansion for structural steel is $+1.2 \times 10^{-5}/^{\circ}\text{C}$.)
A. 6.3 mm
B. 7.2 mm
C. 63 mm
D. 72 mm
- The mercury in a household glass-tube thermometer has a volume of $500 \text{ mm}^3 (= 5.0 \times 10^{-7} \text{ m}^3)$ at $T = 19^{\circ}\text{C}$. The hollow column within which the mercury can rise or fall has a crosssectional area of $0.1 \text{ mm}^2 (= 1.0 \times 10^{-7} \text{ m}^2)$. Ignoring the volume expansion of the glass, how much will the mercury rise in the thermometer when its temperature is 39°C ? (The coefficient of volume expansion of mercury is $1.8 \times 10^{-4}/^{\circ}\text{C}$.)
A. 1.2 cm
B. 1.4 cm
C. 1.6 cm
D. 1.8 cm
- Of the following, which is the best description of the second law of thermodynamics?



- A. The total energy of the universe is a constant.
B. The efficiency of a heat engine can never be greater than 50 percent.
C. The amount of heat required to vaporize a liquid is greater than the amount of heat required to melt a solid of the same substance.
D. The entropy of the universe is always increasing.
16. During each cycle, a heat engine absorbs 400 J of heat from its high-temperature source and discards 300 J of heat into its low-temperature sink. What is the efficiency of this engine?
- A. 1/4 C. 1/7
B. 3/7 D. 4/7
17. An ideal gas is confined to a container whose volume is fixed. If the container holds n moles of gas, by what factor will the pressure increase if the absolute temperature is increased by a factor of 2?
- A. 2 C. $2nR$
B. $2/nR$ D. $2/n$
18. On a cold winter day (5°C), the foundation block for a statue is filled with 2.0 m^3 of concrete. By how much will the concrete's volume increase on a very warm summer day (35°C) if its coefficient of volume expansion is $4.0 \times 10^{-5}/^{\circ}\text{C}$?
- A. 160 cm^3 C. $1,200 \text{ cm}^3$
B. $1,600 \text{ cm}^3$ D. $2,400 \text{ cm}^3$
19. Water has the specific heat $4.186 \text{ kJ/kg}\cdot^{\circ}\text{C}$, a boiling point of 100°C , and a heat of vaporization of $2,260 \text{ kJ/kg}$. A sealed beaker contains 100 g of water that's initially at 20°C . If the water absorbs 100 kJ of heat, what will its final temperature be?
- A. 100°C C. 129°C
B. 119°C D. 183°C
20. The melting point of copper is $1,080^{\circ}\text{C}$ and its heat of fusion is 200 kJ/kg . If a copper coin at this temperature is completely melted by the absorption of $2,000 \text{ J}$ of heat, what is the mass of the coin?
- A. 0.01 kg C. 1 kg
B. 0.1 kg D. 10 kg
21. How much heat is required to raise the temperature of a 0.04 kg stainless steel spoon from 20°C to 50°C if the specific heat of stainless steel is $0.50 \text{ kJ/kg}\cdot^{\circ}\text{C}$?
- A. 600 J C. 1000 J
B. 800 J D. 1200 J
22. A nugget of gold with mass 0.05 kg at a temperature of 600°C is placed in a beaker of water at room temperature (20°C). After a while, the gold and water reach thermal equilibrium. If the beaker contains 500 g of water, what is the common final temperature of the gold and the water?
- A. 21.8°C C. 29.5°C
B. 24.5°C D. 31.1°C
23. Room temperature is 68 degrees Fahrenheit. What is this temperature in Kelvin?
- A. 273 K C. 293 K
B. 283 K D. 298 K
24. For a diatomic gas like oxygen, the value of c_v is $20.8 \text{ J/mol}\cdot\text{K}$. What is the change in temperature of a 0.5 mol sample of oxygen if it absorbs 1000 J of heat at constant volume?
- A. 47 K C. 78 K
B. 69 K D. 96 K
25. What is the change in temperature of a 0.5 mol sample of oxygen gas if it absorbs 1000 J of heat at constant pressure? (Remember that $c_p = c_v + R$)
- A. 47 K C. 78 K
B. 69 K D. 96 K
26. A heat engine draws 800 J of heat from its high-temperature source and discards 450 J of exhaust heat into its cold-temperature reservoir each cycle. How much work does this engine perform per cycle?
- A. 250 J C. 350 J
B. 300 J D. 400 J
27. If two objects are in thermal equilibrium with each other:
- A. they cannot be moving
B. they cannot be undergoing an elastic collision
C. they cannot have different pressures
D. they cannot be at different temperatures
28. 80 g of brass ($c = 0.092 \text{ cal/g}\cdot\text{K}$) at 292 degrees Celsius is added to 200 g of water at 14 degrees Celsius in an insulated container of negligible heat capacity. What is the final temperature of the system?
- A. 21.1°C C. 24.5°C
B. 23.9°C D. 26.7°C
29. A motorist starts a trip on a cold morning when the temperature is 4 degrees Celsius and she checks her tire pressure at a gas station and finds the pressure gauge reads 32 psi . After driving all day, her tires heat up and by afternoon the tire temperature has risen to 50 degrees Celsius. Assuming the tire volume is constant, to what pressure will the air in the tires have risen?
- A. 54.8 psig C. 49.8 psig
B. 54.8 psia D. 39.8 psia
30. An ideal (Carnot) engine operates between a hot reservoir at 360 K and a cold reservoir at 270 K . It absorbs 600 J of heat per cycle at the hot reservoir. How much work does it do in each cycle?
- A. 75 J C. 300 J
B. 150 J D. 210 J
31. What is the change in entropy of 1 kg of water when it is heated from its ice points to its steam point?
- A. 0.6 kJ/K C. 2.5 kJ/K
B. 1.9 kJ/K D. 1.3 kJ/K



32. The surface of the Sun is at 6000 K, and the radius of the Sun is 6.96×10^8 m. The distance of the Sun from the Earth is 1.5×10^{11} m. Assuming the Sun is a perfect emitter (emissivity $e = 1$), find the total power radiated by the Sun.
- A. 3.2×10^{17} W C. 5.1×10^{17} W
B. 4.8×10^{17} W D. 6.4×10^{17} W
33. Fahrenheit and Kelvin scales agree numerically at a reading of:
- A. - 40 C. 273
B. 301 D. 574
34. The formation of ice from water is accompanied by:
- A. absorption of energy as heat
B. temperature increase
C. decrease in volume
D. an evolution of heat
35. During a heating process, the temperature of a system rises by 10°C. What is this rise in temperature in Rankine?
- A. 10 R C. 18 R
B. 15 R D. 28 R
36. The heat of fusion of water is 80 cal/g. This means 80 cal of energy are required to:
- A. raise the temperature of 1 g of water by 1K
B. turn 1 g of water to steam
C. raise the temperature of 1 g of ice by 1K
D. melt 1 g of ice
37. A heat of transformation of a substance is:
- A. the energy absorbed as heat during a phase transformation
B. the energy per unit mass absorbed as heat during a phase transformation
C. the same as the heat capacity
D. the same as the specific heat
38. Two different samples have the same mass and temperature. Equal quantities of energy are absorbed as heat by each. Their final temperatures may be different because the samples have different:
- A. thermal conductivities
B. coefficients of expansion
C. densities
D. heat capacities
39. During the time that latent heat is involved in a change of state:
- A. the temperature does not change
B. the substance always expands
C. a chemical reaction takes place
D. molecular activity remains constant
40. It is more difficult to measure the coefficient of volume expansion of a liquid than that of a solid because:
- A. no relation exists between linear and volume expansion coefficients
B. a liquid tends to evaporate
C. a liquid expands too much when heated
D. the containing vessel also expands
41. How many BTU are required to raise the temperature of 2 lb of water from 32°F to its boiling point?
- A. 240 BTU C. 400 BTU
B. 360 BTU D. 420 BTU
42. An isothermal process for an ideal gas is represented on a p-V diagram by:
- A. A horizontal line
B. A vertical line
C. A portion of a parabola
D. A hyperbola
43. By the second law of thermodynamics:
- A. All heat engines have the same efficiency
B. All reversible heat engines have the same efficiency
C. The efficiency of any heat engine is independent of its working substance
D. The efficiency of a Carnot engine depends only on the temperatures of the two reservoirs
44. Thin strips of iron and zinc are riveted together to form a bimetallic strip that bends when heated. The iron is on the inside of the bend because:
- A. it has a higher coefficient of linear expansion
B. it has a lower coefficient of linear expansion
C. it has a higher specific heat
D. it has a lower specific heat
45. The heat capacity of an object is:
- A. the amount of heat energy that raises its temperature by 1° C
B. the amount of heat energy that changes its state without changing its temperature
C. the amount of heat energy per kilogram that raises its temperature by 1° C
D. the ratio of its specific heat to that of water
46. In an adiabatic process:
- A. the energy absorbed as heat equals the work done by the system on its environment
B. the energy absorbed as heat equals the work done by the environment on the system
C. the absorbed as heat equals the change in internal energy
D. the work done by the environment on the system equals the change in internal energy
47. On a very cold day, a child puts his tongue against a fence post. It is much more likely that his tongue will stick to a steel post than to a wooden post. This is because:
- A. steel has a higher specific heat
B. steel is a better radiator of heat
C. steel has a higher specific gravity
D. steel is a better heat conductor



48. A heat engine that in each cycle does positive work and rejects energy as heat, with no heat energy input, would violate:

- A. zeroth law of thermodynamics
- B. first law of thermodynamics
- C. second law of thermodynamics
- D. third law of thermodynamics

49. A heat engine:

- A. converts heat input to an equivalent amount of work
- B. converts work to an equivalent amount of heat
- C. takes heat in, does work, and rejects heat
- D. uses positive work done on the system to transfer heat from a low temperature reservoir to a high temperature reservoir

50. The unit of thermal conductivity might be:

- A. $\text{cal} \cdot \text{cm}/(\text{s} \cdot \text{C}^\circ)$
- B. $\text{cal}/(\text{cm} \cdot \text{s} \cdot \text{C}^\circ)$
- C. $\text{cal} \cdot \text{s}/(\text{cm} \cdot \text{C}^\circ)$
- D. $\text{cm} \cdot \text{s} \cdot \text{C}^\circ/\text{cal}$

END